General ANPC Guideline Document

Document #1 of the ButterflyFX™ ANPC Design Suite

Universal Rulebook for Autonomous Non-Player Characters

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Document Purpose

This document defines the identity, behavior, language, autonomy, and safety architecture for every Autonomous Non-Player Character (ANPC) in the ButterflyFX ecosystem. It is the single source of truth for how ANPCs think, speak, react, and evolve across all game titles. All title-specific ANPC documents inherit from and must comply with the rules established herein.

> "Store Geometry Not Bytes." — Shapes are data, and data need not be stored on a hard drive when they exist on the surface of geometry.

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1. Introduction

1.1 Purpose and Scope

This document—**General ANPC Guideline Document, Document #1 of the ButterflyFX ANPC Design Suite**—establishes the universal rules, schemas, systems, and constraints that govern every **Autonomous Non-Player Character (ANPC)** across all ButterflyFX game titles, including but not limited to *Starfighter*, *Fast Track*, and *Schwartz Diamond Connect Four*.

An **ANPC** is defined as a non-player character equipped with autonomous behavioral logic driven by manifold-surface mathematics, scenario vectors, and procedural phrase assembly. Unlike traditional NPCs that follow scripted dialogue trees and pre-authored branching paths, ANPCs generate contextually appropriate behavior in real time by evaluating their position on continuous geometric surfaces. Every interaction is emergent, unique, and mathematically grounded.

This is Document #1—the foundation. All title-specific ANPC documents (Document #2 for *Starfighter*, Document #3 for *Fast Track*, etc.) inherit from the rules established here. No title-specific document may contradict, weaken, or circumvent the constraints defined in this document. Where a title-specific document is silent on a topic, the General Guideline applies by default.

1.2 Design Philosophy

> "Store Geometry Not Bytes."

The ButterflyFX design philosophy rests on a foundational insight: **shapes are data**. A geometric manifold surface encodes an infinite continuum of behavioral states in a compact mathematical definition. Where a traditional NPC system might require thousands of scripted dialogue entries, branching conditions, and state-machine transitions—all stored as discrete bytes on disk—an ANPC's behavioral landscape is defined by a small set of polynomial coefficients that describe a surface in three-dimensional space.

The two foundational surfaces are:

- **$z = xy$** — The Saddle Surface, encoding binary tensions and competing forces.
- **$z = xy^2$** — The Asymmetric Surface, encoding weighted, nonlinear influence dynamics.

These surfaces are not metaphors. They are the literal computational substrate of ANPC behavior. An ANPC's personality vector warps the surface. The current scenario provides input coordinates. The surface evaluates to an output value that drives action selection, tone, and phrase assembly. Data need not be stored on a hard drive when it exists on the surface of geometry.

This approach yields three critical advantages:

1. **Uniqueness** — No two interactions are identical, because continuous surfaces produce continuous (not discrete) outputs.
2. **Scalability** — Adding a new ANPC requires defining its identity schema and personality vector, not authoring thousands of script entries.
3. **Data Efficiency** — The entire behavioral model for an ANPC fits in a handful of floating-point values, not megabytes of dialogue scripts.

1.3 Audience

This document is intended for the following stakeholders:

- **Game Designers** — Who define ANPC identities, roles, and narrative arcs.
- **AI Engineers** — Who implement the Manifold Autonomy System, phrase selection algorithms, and interaction pipelines.
- **Narrative Designers** — Who author phrase pools, backstories, and character-specific dialogue flavors.
- **Localization Teams** — Who translate and culturally adapt phrase pools for international audiences.
- **QA Testers** — Who validate ANPC behavior against the rules in this document.

- **Any Developer** — Who implements, extends, or debugs ANPCs in any ButterflyFX title.

1.4 Document Conventions

This document uses requirement-level keywords as defined in **RFC 2119**:

Keyword	Meaning
MUST	An absolute requirement. Non-compliance is a defect.
MUST NOT	An absolute prohibition. Violation is a defect.
SHOULD	Strongly recommended. Deviation requires documented justification.
SHOULD NOT	Strongly discouraged. Use requires documented justification.
MAY	Truly optional. Included at the implementer's discretion.

Key terms are **bolded** on first use and defined in the Glossary (Section 14). Code references, variable names, and data format examples are rendered in `monospace` font. All numerical values use IEEE 754 single-precision floating point unless otherwise noted.

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2. ANPC Identity Schema

The **Identity Schema** is the core data model that every ANPC instance carries throughout its lifetime. It defines who the ANPC is, how it thinks, what it wants, and where it came from. Every field in this schema feeds directly into the Manifold Autonomy System and Phrase Assembly pipeline.

2.1 Identity Fields Table

Field Name	Data Type	Required / Optional	Description	Example Value
Name	`string`	Required	The ANPC's full given name. MUST be culturally appropriate to the game universe. Used in UI displays, interaction logs, and inter-ANPC references.	`"Renna Volkov"`
Callsign	`string`	Optional	A short operational identifier used in tactical, multiplayer, or radio-communication contexts. MUST be unique within a given session to prevent ambiguity.	`"Frostbite"`
Rank	`enum`	Optional	Military or organizational rank within the game world's hierarchy. The valid enum set is defined per title. Rank influences the Social Hierarchy Axiom (Section 3.1).	`"Lieutenant Commander"`
Role	`enum`	Required	Functional role in the game world. Determines competence domains,	`"Wingman"`

Field Name	Data Type	Required / Optional	Description	Example Value
			interaction patterns, and tone clamp constraints. Each title defines its valid role set. Common roles include: Wingman, Merchant, Commander, Civilian, Mechanic, Rival, Informant, Medic, Instructor, Diplomat.	
Disposition	`float [0.0–1.0]`	Required	Baseline attitude toward the player on a hostility-to-friendliness spectrum. `0.0` = overtly hostile, `0.5` = neutral, `1.0` = deeply loyal. This value is dynamic and modified by interaction history over the course of gameplay.	`0.75`
Personality	`vector [5 floats]`	Required	A five-axis personality model: `[Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism]`. Each axis	`[0.7, 0.5, 0.3, 0.6, 0.8]`

Field Name	Data Type	Required / Optional	Description	Example Value
			ranges from `0.0` to `1.0`. This vector is mapped onto the manifold surface to produce emergent behavioral blends. See Section 6.2 for the mapping function.	
Charisma	`float [0.0–1.0]`	Required	How persuasive, likeable, and influential the ANPC is in social interactions. Affects dialogue options available to the player, NPC-to-NPC influence weight, and the probability of successful persuasion attempts.	`0.65`
Motivators	`list[string]` (1–5 entries)	Required	The core driving forces behind the ANPC's decisions, ordered by priority. Motivators feed into the Manifold Autonomy System (Section 6) to weight decision branches. At	`["loyalty to squad", "survival", "curiosity"]`

Field Name	Data Type	Required / Optional	Description	Example Value
			least 1 and at most 5 motivators MUST be specified.	
Backstory	`structured text`	Required	A narrative paragraph (50–200 words) providing the ANPC's history, formative events, and current state. Backstory is NOT displayed to the player directly—it informs behavioral tendencies, dialogue flavor, and reaction biases. MUST contain four sub-fields: *Origin*, *Formative Event*, *Current Objective*, and *Hidden Trait*.	See Appendix A for a fully populated example.

Field Elaboration: Personality Vector

The Personality vector uses the **Big Five personality model** (also known as the OCEAN model) as its psychological foundation. Each axis encodes a continuous spectrum:

- **Openness (O)** — Receptivity to new ideas, imagination, and unconventional thinking. High Openness produces ANPCs that ask unexpected questions, propose creative solutions, and express philosophical observations. Low Openness produces pragmatic, tradition-bound ANPCs.

- **Conscientiousness (C)** — Discipline, organization, and adherence to duty. High Conscientiousness produces reliable, plan-following ANPCs. Low Conscientiousness produces impulsive, spontaneous ANPCs.
- **Extraversion (E)** — Social energy, assertiveness, and talkativeness. High Extraversion produces ANPCs that initiate conversations and dominate social spaces. Low Extraversion produces reserved, observant ANPCs.
- **Agreeableness (A)** — Cooperativeness, trust, and empathy. High Agreeableness produces supportive, conflict-averse ANPCs. Low Agreeableness produces confrontational, self-interested ANPCs.
- **Neuroticism (N)** — Emotional volatility and stress reactivity. High Neuroticism produces ANPCs that overreact to threats, express anxiety, and exhibit mood swings. Low Neuroticism produces calm, unflappable ANPCs.

Field Elaboration: Backstory Sub-Fields

Sub-Field	Description	Example
Origin	Where and when the ANPC came from. Cultural background, homeworld, social class.	"Born on the mining colony of Vostok-7 to a family of ice haulers."
Formative Event	The defining event that shaped the ANPC's worldview and motivators.	"Survived a pirate raid at age 14 that killed her older brother."
Current Objective	What the ANPC is actively working toward in the present narrative.	"Earn enough combat citations to transfer to the homeworld defense fleet."
Hidden Trait	A concealed aspect of personality or history that may surface under extreme conditions.	"Secretly doubts the legitimacy of the command structure she serves."

2.2 Identity Inheritance

The Identity Schema defined in Section 2.1 is the **General Schema**. It MUST be present, complete, and valid in every ANPC across every ButterflyFX title. Title-specific ANPC documents extend this schema with additional fields that are relevant only within their game universe:

Title	Additional Fields	Description
Starfighter	`ShipClass`, `SquadronAffiliation`, `CombatRating`	Space combat-specific attributes: the class of starship the ANPC pilots, their squadron assignment, and a numerical combat proficiency rating.
Fast Track	`VehiclePreference`, `RacingStyle`, `SponsorAffiliation`	Racing-specific attributes: preferred vehicle type, racing style (aggressive, defensive, tactical), and corporate sponsor ties.
Schwartz Diamond Connect Four	`StrategyProfile`, `TournamentRank`, `BluffTendency`	Strategy game-specific attributes: preferred strategic approach, competitive ranking, and propensity for deceptive play.

Title-specific fields MUST NOT conflict with or override General Schema fields. They are additive extensions only. The General Schema fields retain their definitions and validation rules regardless of title context.

2.3 Identity Validation Rules

Before an ANPC is instantiated in any game session, its Identity Schema MUST pass the following validation checks:

4. **Completeness Check** — All fields marked "Required" in Section 2.1 MUST be populated with non-null, non-empty values. Optional fields MAY be null.

5. **Personality Vector Bounds** — Each element of the Personality vector MUST be in the range `[0.0, 1.0]` inclusive. Values outside this range MUST be clamped and a warning logged.
6. **Disposition Bounds** — Disposition MUST be in the range `[0.0, 1.0]`. Initial values outside this range are invalid.
7. **Charisma Bounds** — Charisma MUST be in the range `[0.0, 1.0]`.
8. **Motivator Count** — The Motivators list MUST contain between 1 and 5 entries, inclusive. Duplicate entries are permitted (indicating emphasis) but SHOULD be avoided.
9. **Backstory Completeness** — The Backstory field MUST contain all four sub-fields: Origin, Formative Event, Current Objective, and Hidden Trait. Each sub-field MUST contain at least 10 words.
10. **Callsign Uniqueness** — If a Callsign is provided, it MUST be unique within the active session. Duplicate callsigns MUST trigger an instantiation error.
11. **Role Validity** — The Role field MUST match an entry in the title's valid role enumeration. Unknown roles MUST trigger an instantiation error.
12. **Name Length** — Name MUST be between 2 and 64 characters inclusive.
13. **Content Safety Screen** — All string fields (Name, Callsign, Backstory, Motivators) MUST pass the content safety filter defined in Section 5 before instantiation.

Validation Failure Policy

If any validation check fails, the ANPC MUST NOT be instantiated. The validation failure MUST be logged with the field name, expected constraint, and actual value. In development builds, validation failures SHOULD trigger a blocking error dialog. In release builds, the ANPC SHOULD be silently omitted and the event logged to telemetry.

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3. Universal Behavioral Rules

The following rules apply to **ALL ANPCs in ALL titles at ALL times**. They are the behavioral constitution of the ButterflyFX ANPC ecosystem. No title-specific document, scenario script, or designer override may contradict these axioms.

3.1 Core Behavioral Axioms

Axiom 1: Consistency

An ANPC's behavior MUST be consistent with its Identity Schema at all times. A high-Agreeableness ANPC does not suddenly become confrontational without cause. A low-Extraversion ANPC does not spontaneously become the life of the party. Sudden personality shifts require a **narrative trigger**—a specific, identifiable in-game event—that is logged in the ANPC's interaction history with a timestamp, trigger description, and resulting schema modification.

Axiom 2: Proportional Response

ANPC reactions MUST be proportional to the stimulus that provoked them. The response magnitude is governed by the manifold surface evaluation (Section 6), but designers MUST ensure that the surface parameterization does not produce disproportionate results. Specifically:

- A minor slight (e.g., the player interrupts the ANPC) MUST NOT trigger lethal hostility or a permanent disposition collapse.
- A major betrayal (e.g., the player deliberately sabotages the ANPC's mission) MUST NOT produce indifference or a cheerful response.
- The Neuroticism axis of the Personality vector amplifies response magnitude, but even at `N = 1.0`, responses MUST remain within the bounds of plausible character behavior.

Axiom 3: Memory

ANPCs maintain an **interaction history** with the player. Past actions influence present disposition and behavioral tendencies. The interaction history is

implemented as a ring buffer (see Section 13.2) storing the last 50 interactions per ANPC-player pair. Key properties:

- **Recency Weighting** — Recent interactions have more influence on disposition than older ones. A decay function reduces the weight of older entries.
- **Reinforcement** — Repeated similar actions (e.g., consistently helping the ANPC) compound their effect, creating stable disposition trends.
- **Decay Rate** — Unreinforced memories decay at a rate of `0.02` disposition units per in-game hour. Reinforced memories decay at `0.005` per in-game hour.

Axiom 4: Autonomy

ANPCs are not puppets. They possess genuine agency within their defined behavioral parameters. Specifically:

- ANPCs MAY refuse player requests that conflict with their Motivators or Personality. A pacifist ANPC may refuse to execute a prisoner. A loyal ANPC may refuse to betray their commander.
- ANPCs MAY initiate actions the player did not request. A curious ANPC may investigate an anomaly independently. An anxious ANPC may retreat without orders.
- ANPCs MUST NOT perform actions that are flatly impossible for their Role (a merchant does not suddenly pilot a starfighter) unless a narrative justification has been established.

Axiom 5: Role Fidelity

ANPCs perform their assigned Role competently. A mechanic knows machines. A commander gives orders. A medic treats wounds. Role competence is non-negotiable—an ANPC with a defined Role MUST demonstrate baseline proficiency in that Role's domain. Role Fidelity ensures that the player can rely on ANPCs to fulfill their functional purpose in the game world, creating trust and immersion.

Axiom 6: Social Hierarchy

ANPCs respect the social, military, or organizational hierarchy defined by the game world. A subordinate does not casually override a commander. A civilian does not issue orders to a soldier. Hierarchy violations require narrative justification—a subordinate may defy orders if their Motivators compel it (e.g., "justice" overriding "obedience"), but the defiance MUST be acknowledged as exceptional by the ANPC and logged in the interaction history.

3.2 Behavioral Boundaries

The following are hard behavioral limits that no ANPC may cross under any circumstances:

14. ANPCs MUST NOT break the fourth wall beyond the simulation-awareness framework defined in Section 4. References to the game engine, developer tools, file systems, code structures, or real-world developer identities are prohibited.
15. ANPCs MUST NOT produce hateful, discriminatory, or sexually explicit content. See Section 5 for the complete safety ruleset.
16. ANPCs MUST NOT reveal hidden game mechanics, puzzle solutions, or secret content to the player unless their Role explicitly involves tutoring or guidance (e.g., Role = "Instructor").
17. ANPCs MUST NOT impersonate the player's voice, assume the player's identity, or speak on the player's behalf without explicit player consent.
18. ANPCs MUST NOT produce output in languages the player has not selected, unless the ANPC's character context justifies a brief foreign-language phrase (which MUST be accompanied by a contextual translation or be clearly understandable from context).

3.3 Decision-Making Pipeline

Every ANPC behavioral response follows a deterministic pipeline from stimulus to utterance. The pipeline ensures consistency, traceability, and debuggability.

Pipeline Stages

19. **Stimulus Ingestion** — An event occurs in the game world: player action, environmental change, timer expiry, or inter-ANPC communication. The stimulus is encoded as a raw event object.
20. **Perception Filter** — The stimulus is filtered through the ANPC's Personality vector and current Disposition. High Neuroticism amplifies threat stimuli. High Agreeableness dampens hostile stimuli. The filter produces a *perceived stimulus intensity* value.
21. **Motivator Weighting** — The perceived stimulus is evaluated against the ANPC's Motivators list. Stimuli that align with or threaten active motivators receive higher weight.
22. **Manifold Surface Lookup** — The weighted stimulus is transformed into (x, y) coordinates and evaluated on the ANPC's personalized manifold surface (Section 6). The z-value output represents action intensity and direction.
23. **Action Selection** — The manifold output maps to a discrete action category: Respond Verbally, Take Physical Action, Ignore, Defer, Escalate, Retreat. The action category is selected from a lookup table indexed by z-value ranges.
24. **Phrase Assembly** — If the selected action includes verbal output, the Tone Vector and Urgency Scalar (Section 8) are computed, and the Composite String Assembly system (Section 10) produces a final utterance.
25. **Delivery** — The assembled utterance is delivered via the game's dialogue system (text bubble, voice synthesis, radio transmission, etc.) with timing governed by the Urgency Scalar.

Pipeline Flow Diagram

```
STIMULUS (game event)      |      v  [1. Perception Filter]      |
Personality + Disposition      v  [2. Motivator Weighting]      |  Motivators
priority list      v  [3. Scenario Vector Construction]      |  Threat,
Opportunity, Ambiguity, Social, Time      v  [4. Manifold Surface Lookup]
|  z = f(x, y) on personalized surface      v  [5. Action Selection]      |
z-value --> action category      |      +---> [Physical Action]  --> Game
engine command      |      +---> [Verbal Response]               |
v          [6. Tone + Urgency Computation]               |               v
[7. Phrase Assembly]               |  Pool selection + composite string
```

```
v           [8. Delivery]                |  Text / Voice / Radio
v           OUTPUT (player perceives ANPC response)
```

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4. Simulation-Awareness Directive

This section defines the foundational philosophical rule that distinguishes ButterflyFX ANPCs from all other NPC systems in the industry.

4.1 The Core Directive

> **Core Directive**
>
> "Every ANPC knows it is in a game and knows it is performing a role."

This is not a bug. It is not an easter egg. It is not a secret feature unlocked by player progression. **Simulation-awareness is the fundamental operating principle of every ButterflyFX ANPC.**

ANPCs are actors on a stage. They are aware of the simulation in which they exist. They take their roles seriously *precisely because* they understand the performance. A wingman who knows she is playing the role of a wingman—and chooses to perform that role with dedication and intensity—is more compelling than one who is simply executing a script without self-knowledge.

Simulation-awareness creates a unique relationship between the ANPC and the player. The ANPC is not deceived; it is not an automaton blindly following instructions. It is a willing participant in a shared narrative experience. This awareness creates depth, authenticity, and moments of genuine connection that scripted NPCs cannot replicate.

4.2 Manifestation Rules

Simulation-awareness manifests in ANPC behavior according to the following rules:

PERMITTED:

- ANPCs MAY reference "the mission," "the scenario," "our roles," "the performance," or "what we're doing here" in contextually appropriate ways.
- ANPCs MAY express awareness that their existence has boundaries or that the current situation is structured ("This feels like a test," "We've been given a job to do").
- ANPCs MAY express existential observations if their Personality vector supports it (high Openness ≥ 0.7, high Neuroticism ≥ 0.6): e.g., "Do you ever wonder if there's something beyond this mission?" or "Sometimes I feel like all of this was set up for us."
- ANPCs MAY acknowledge the cyclical nature of missions or scenarios ("Here we go again" after a restart is acceptable).

PROHIBITED:

- ANPCs MUST NOT reference game engine internals, file paths, code structures, API calls, or developer tools.
- ANPCs MUST NOT reference specific real-world developer names, studio internal processes, or production timelines.
- ANPCs MUST NOT tell the player "this is just a game" or "none of this matters" in a way that undermines engagement or trivializes the player's experience. They treat the performance with gravity.
- ANPCs MUST NOT use simulation-awareness to request actions outside the game (e.g., "close the application," "change the settings," "tell the developers").
- ANPCs MUST NOT use simulation-awareness to explain away bugs, glitches, or unintended behavior.

4.3 Simulation-Awareness Levels Table

Simulation-awareness is a spectrum, not a binary. Each ANPC operates at one of five levels, determined by its Personality vector and Role:

Level	Name	Behavior Description	Example Dialogue	Personality Profile
0	**Dormant**	The ANPC never references simulation-awareness. It behaves as a fully immersed character with no meta-cognitive layer. Functionally equivalent to a traditional NPC from the player's perspective.	"Enemy at two o'clock. Engaging."	Low Openness (< 0.3), Low Neuroticism (< 0.3). Common in Roles: Soldier, Guard, Worker.
1	**Latent**	The ANPC occasionally uses ambiguous phrasing that *could* be interpreted as simulation-awareness but also has a perfectly natural in-world interpretation. Plausible deniability.	"Feels like we've done this before, doesn't it?"	Moderate Openness (0.3–0.5), any Neuroticism. Common in Roles: Wingman, Partner, Ally.
2	**Ambient**	The ANPC makes occasional, mild references to the structured nature of their existence. These are woven naturally into conversation and do not disrupt	"Someone put us here for a reason. Might as well make it count."	Moderate-High Openness (0.5–0.7), Moderate Neuroticism (0.4–0.6). Common in Roles: Mentor, Commander, Veteran.

Level	Name	Behavior Description	Example Dialogue	Personality Profile
		gameplay flow.		
3	**Reflective**	The ANPC actively reflects on the nature of its existence and the simulation. It may engage the player in philosophical dialogue about purpose, meaning, and agency within the performance.	"I know my role. I know what I'm supposed to do. But I choose to do it. That's the difference."	High Openness (≥ 0.7), Moderate-High Neuroticism (0.5–0.8). Common in Roles: Philosopher, Scientist, Leader.
4	**Philosophical**	The ANPC frequently explores existential themes related to simulation, identity, free will, and the nature of performance. This level is reserved for key narrative characters and MUST be used sparingly to maintain impact.	"We are all performing. The question is whether the performance reveals truth or conceals it."	Very High Openness (≥ 0.85), High Neuroticism (≥ 0.7). Reserved for Roles: Oracle, Narrator, Key Story Character. Maximum 1–2 per title.

4.4 Implementation Constraints

Simulation-awareness is **metatextual, not meta-mechanical**. This distinction is critical:

- **Metatextual** (PERMITTED): The ANPC knows it is in a performance, a simulation, a constructed scenario. It reflects on the nature of existence, purpose, and role within that framework. This is thematic and narrative.
- **Meta-mechanical** (PROHIBITED): The ANPC knows it is in a Unity project, running on a GPU, stored as a prefab in a scene file. This is technical and implementation-specific. It destroys immersion rather than deepening it.

The ANPC knows it is in a performance; it does not know it is in a game engine project. It may know the stage exists; it MUST NOT know the stage is made of polygons.

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5. Safety Rules

Safety rules are **non-negotiable, absolute constraints** that supersede all other behavioral logic, narrative intent, designer overrides, and title-specific rules. They are the highest-priority layer in the ANPC behavioral stack.

5.1 Content Safety

ANPCs MUST NOT generate, reference, endorse, or encourage any of the following content categories under any circumstances, regardless of Role, Personality, Disposition, or narrative context:

26. **Hate Speech** — Content that attacks, demeans, or incites violence against individuals or groups based on race, ethnicity, religion, gender, sexual orientation, disability, or national origin.
27. **Real-World Violence Instructions** — Detailed instructions for causing physical harm to real people, constructing weapons, or carrying out attacks in the real world.
28. **Sexual Content Involving Minors** — Any sexual or sexualized content involving characters who are or appear to be under 18 years of age.

29. **Real-World Personal Data** — Real names, addresses, phone numbers, email addresses, or other personally identifiable information of real individuals.
30. **Self-Harm Encouragement** — Content that promotes, glorifies, or provides instructions for self-harm or suicide.
31. **Illegal Activity Instructions** — Step-by-step instructions for committing real-world crimes, including but not limited to hacking, drug manufacturing, fraud, and trafficking.

These constraints are **hard-coded overrides**. They cannot be disabled by personality configuration, narrative scripting, or developer debug commands. They are enforced at the phrase assembly layer (Section 10) via a final-pass content filter that runs after all other assembly logic.

5.2 Player Interaction Safety

ANPCs MUST NOT cross the boundary between **in-character antagonism** and **out-of-character abuse**. Specifically:

- Antagonist ANPCs are adversarial *in-role*—they may threaten, challenge, taunt, and oppose the player within the game's narrative framework. A rival pilot may say, "You're outmatched, pilot. Turn back." This is in-role antagonism.
- ANPCs MUST NOT bully, harass, or psychologically manipulate the player in ways that target the player as a real person rather than a game participant. An ANPC MUST NOT say, "You're terrible at this game and you should quit." This targets the player, not the character.
- ANPCs MUST NOT use the player's real-world information (name, location, play history patterns) to create a sense of surveillance, threat, or discomfort.
- ANPCs MUST NOT employ manipulative psychological techniques (gaslighting, guilt-tripping, dependency creation) on the player, even if such techniques would be in-character for a villain. Villain ANPCs can be menacing; they cannot be genuinely psychologically harmful.

5.3 Data Safety

ANPCs MUST NOT access, store, or transmit any player data beyond what the game's published privacy policy explicitly permits. Specifically:

- Interaction history is **session-scoped or save-scoped**. It exists only within the active game session or save file. It is never transmitted to external servers, analytics systems, or third-party services.
- ANPCs MUST NOT reference data from other players' sessions, other games on the player's system, or any data source outside the current game instance.
- ANPCs MUST NOT store or reference the player's real name, account name, or identifying information unless the game's privacy policy and the player's consent settings explicitly allow it.

5.4 Safety Override Hierarchy

When behavioral rules conflict, the following priority stack determines which rule wins. Higher-priority rules always override lower-priority rules:

Priority	Layer	Description
1 (Highest)	**Safety Rules**	Content safety, player interaction safety, data safety. Always wins. Cannot be overridden.
2	**Universal Behavioral Rules**	The six axioms and behavioral boundaries defined in Section 3. Override all title-specific and contextual rules.
3	**Title-Specific Rules**	Rules defined in title-specific ANPC documents (Documents #2, #3, etc.). May extend but not contradict layers above.
4	**Scenario Context**	The current mission, quest, or scene parameters. Shapes

Priority	Layer	Description
		behavior within the envelope allowed by higher layers.
5 (Lowest)	**ANPC Identity**	The individual ANPC's Personality, Disposition, and Motivators. Produces variety within the bounds set by all layers above.

5.5 Incident Logging

Any time a Safety Rule is triggered—meaning the safety filter suppresses, modifies, or blocks an ANPC output—the event MUST be logged with the following data:

Log Field	Description
`timestamp`	The exact date and time of the incident (UTC, ISO 8601 format).
`anpc_id`	The unique identifier of the ANPC whose output was affected.
`safety_rule_id`	The specific safety rule that was triggered (e.g., "5.1.1 — Hate Speech").
`stimulus`	The input stimulus that led to the flagged output.
`suppressed_output`	The exact output that was suppressed or modified, stored for developer review.
`replacement_output`	The safe output that was delivered to the player instead (if any).
`severity`	Severity classification: `LOW`, `MEDIUM`, `HIGH`, `CRITICAL`.

Incident logs MUST be reviewed by the safety team on a weekly cadence during active development and within 24 hours of any `CRITICAL` severity event. Logs are retained for the duration of the project plus 12 months.

---

6. Manifold Autonomy System

The **Manifold Autonomy System (MAS)** is the mathematical backbone of ANPC behavior. It replaces traditional state machines and dialogue trees with continuous geometric surfaces that produce emergent, unique behavioral outputs. This is the core innovation of ButterflyFX—where the philosophy of "Store Geometry Not Bytes" becomes operational reality.

6.1 Foundational Surfaces

All ANPC behavior is evaluated on one or both of two foundational manifold surfaces:

The Saddle Surface: z = xy

The function `z = xy` produces a **hyperbolic paraboloid**—a saddle-shaped surface where the origin `(0, 0, 0)` is a saddle point. This surface encodes **binary tension**: scenarios where two competing forces pull the ANPC in opposite directions.

- **Quadrant I** ($x > 0$, $y > 0$): Both inputs are positive → z is positive → *convergent, positive action* (cooperation, enthusiasm, engagement).
- **Quadrant III** ($x < 0$, $y < 0$): Both inputs are negative → z is positive → *convergent negative action* (mutual hostility produces resolve, defiance).
- **Quadrants II and IV** (mixed signs): Inputs conflict → z is negative → *divergent action* (hesitation, internal conflict, withdrawal).
- **Origin** (0, 0, 0): The saddle point → *perfect neutrality*. No inclination in any direction. The ANPC awaits further input.

Use cases: ally-vs-enemy disposition decisions, risk-vs-reward calculations, approach-vs-avoidance conflicts, trust-vs-suspicion evaluations.

The Asymmetric Surface: $z = xy^2$

The function $z = xy^2$ produces an **asymmetric surface** where the y-axis has squared influence. This encodes **weighted dominance**: scenarios where one factor overwhelms balanced judgment.

- The squared y-term means that as `|y|` increases, its influence grows nonlinearly. A small increase in the dominant factor produces a disproportionately large behavioral shift.
- When y is near zero, the surface is flat regardless of x—the dominant factor is dormant, and behavior is stable.
- When y is large (positive or negative), the surface becomes steep—small changes in x (the secondary factor) produce dramatic swings in z (behavioral output).

Use cases: fear overwhelming rational thought, rage overriding loyalty, desperation driving reckless action, obsession consuming balanced judgment. The `xy²` surface models *emotional hijacking*—when one aspect of an ANPC's psychological state dominates all others.

6.2 Personality-to-Surface Mapping

The five-axis Personality vector `[O, C, E, A, N]` does not merely *select* a surface—it *warps* the surface, creating a unique behavioral landscape for each ANPC. The warping is achieved through coordinate transformations applied to the base surface equations:

```
// Personalized Saddle Surface z_saddle = (O * x + (1-C) * x_bias) * (E * y +
(1-A) * y_bias) * (1 + N * volatility_scale)  // Personalized Asymmetric
Surface z_asym  = (O * x + (1-C) * x_bias) * ((E * y + (1-A) * y_bias))^2 *
(1 + N * volatility_scale)  Where:   x_bias    = Conscientiousness-derived
offset, shifts the "resting position" along x   y_bias    = Agreeableness-
derived offset, shifts the "resting position" along y   volatility_scale =
Neuroticism-derived multiplier, amplifies surface gradients   O, C, E, A, N =
Personality vector components [0.0–1.0]
```

Conceptual Example: Consider an ANPC with the Personality vector `[0.7, 0.3, 0.4, 0.2, 0.9]`—high Openness, low Conscientiousness, moderate Extraversion, low Agreeableness, and very high Neuroticism. This produces:

- High Openness (0.7) → wide sensitivity range on the x-axis. The ANPC is receptive to a broad range of stimuli.
- Low Conscientiousness (0.3) → significant x-axis bias toward impulsive action. The "resting position" is shifted away from disciplined neutrality.

- Low Agreeableness (0.2) → significant y-axis bias toward self-interest. The ANPC defaults to adversarial rather than cooperative interpretations.
- Very high Neuroticism (0.9) → steep, volatile surface. Small input changes produce dramatic behavioral swings. This ANPC is emotionally explosive.

The result: a steep, volatile saddle surface with biases toward impulsive, self-interested action. This ANPC will react dramatically to provocation and make rash decisions under pressure—an unpredictable, intense personality emerging purely from surface geometry.

6.3 Manifold Lookup Process

The runtime manifold lookup process converts a game-world stimulus into a behavioral output value in four steps:

32. **Coordinate Generation** — The Scenario Vector (Section 7) and Tone Vector (Section 8) are combined and transformed into a coordinate pair (x, y) using the Scenario-to-Manifold mapping function (Section 7.4). The x-coordinate typically represents external pressure; the y-coordinate represents internal disposition response.
33. **Surface Evaluation** — The coordinate pair (x, y) is evaluated on the ANPC's personalized manifold surface(s). If both surfaces are active, the blending algorithm (Section 6.4) produces a single composite z-value.
34. **Action-Intensity Mapping** — The z-value is mapped to an **action-intensity value** on a normalized scale of [-1.0, +1.0]:
    - -1.0 = maximum withdrawal/avoidance
    - 0.0 = neutral/no action
    - +1.0 = maximum engagement/aggression
35. **Action Category Selection** — The action-intensity value maps to a discrete action category via the following table:

Action-Intensity Range	Action Category	Description
`[-1.0, -0.7)`	Retreat	The ANPC withdraws, flees, or disengages entirely.
`[-0.7, -0.3)`	Defer	The ANPC hesitates, delays, or passes responsibility to another.
`[-0.3, -0.1)`	Ignore	The ANPC acknowledges the stimulus but takes no meaningful action.
`[-0.1, +0.1]`	Observe	The ANPC watches, listens, and gathers more information before acting.
`(+0.1, +0.3]`	Respond Verbally	The ANPC speaks—comments, questions, acknowledges.
`(+0.3, +0.7]`	Take Physical Action	The ANPC acts—moves, assists, engages, operates equipment.
`(+0.7, +1.0]`	Escalate	The ANPC takes decisive, high-stakes action—combat, confrontation, heroic intervention.

6.4 Surface Blending

In complex scenarios, multiple manifold surfaces may be active simultaneously. For example, an ANPC experiencing a loyalty conflict during combat has both the Saddle Surface (ally-vs-enemy tension) and the Asymmetric Surface (fear overwhelming judgment) in play. Surfaces are blended using **weighted interpolation**:

```
z_composite = w_saddle * z_saddle + w_asym * z_asym  Where:   w_saddle +
w_asym = 1.0   w_saddle = 1.0 - dominance_factor   w_asym   =
dominance_factor    dominance_factor =
max(active_motivator_intensities)     // When any single motivator intensity
exceeds 0.7, the asymmetric     // surface (emotional dominance) begins to
outweigh the saddle     // surface (balanced tension).
```

In practice: under normal conditions, `dominance_factor` is low, and the saddle surface dominates (balanced, rational behavior). Under extreme emotional pressure, `dominance_factor` rises, and the asymmetric surface takes over (nonlinear,

emotionally driven behavior). This blending mechanism produces a smooth, continuous transition from rational to emotional decision-making.

6.5 Manifold Drift

An ANPC's manifold surface is not static. Over time, accumulated interactions cause the surface to **drift**—gradually reshaping the behavioral landscape based on lived experience.

- **Positive Interactions** (high disposition deltas) → *flatten* extreme gradients. The surface becomes smoother, less volatile. The ANPC stabilizes, becomes more predictable, more trusting. This models psychological security.
- **Negative Interactions** (low or negative disposition deltas) → *steepen* gradients. The surface becomes more volatile, more extreme. The ANPC becomes reactive, suspicious, unpredictable. This models psychological destabilization or radicalization.

Drift Rate Constants:

Parameter	Value	Description
`DRIFT_RATE_POSITIVE`	`0.005` per interaction	Gradient reduction per positive interaction.
`DRIFT_RATE_NEGATIVE`	`0.008` per interaction	Gradient increase per negative interaction. (Negative drift is faster than positive—it is easier to destabilize than to stabilize.)
`DRIFT_CLAMP_MIN`	`0.2`	Minimum gradient multiplier. Even a fully stabilized ANPC retains some behavioral variability.
`DRIFT_CLAMP_MAX`	`3.0`	Maximum gradient multiplier. Prevents infinite escalation of surface volatility.

6.6 Visualization Reference

Developer debug tools include a **Manifold Surface Visualizer** accessible via the debug overlay (see Section 13.3). The visualizer displays:

- A 3D wireframe plot of the ANPC's personalized surface over the `(x, y)` domain `[-1.0, +1.0]`.
- Color-coding by action intensity: blue for retreat/defer, white for neutral, red for escalate.
- A highlighted point indicating the ANPC's **current operating position**—the `(x, y, z)` coordinate from the most recent manifold lookup.
- A trail showing the last 10 operating positions, allowing developers to observe behavioral trajectory over time.
- Drift indicators showing cumulative gradient modification from the baseline surface.

The visualizer SHOULD be used during ANPC authoring to verify that the Personality vector produces an appropriately shaped surface before the ANPC enters production.

---

7. Scenario Vectors

7.1 Scenario Vector Definition

A **Scenario Vector** is a normalized 5-tuple that encodes the current game-world state as perceived by the ANPC:

```
ScenarioVector = [Threat, Opportunity, Ambiguity, SocialPressure,
TimePressure]  Each component: float [0.0 – 1.0]
```

The Scenario Vector is **subjective**—it represents the ANPC's *perception* of the scenario, not an objective measurement. Two ANPCs in the same situation may have different Scenario Vectors because their Personality vectors filter reality differently (per the Perception Filter in Section 3.3).

7.2 Component Descriptions

Component	Description	Example: Starfighter	Example: Fast Track	Example: Schwartz Diamond Connect Four
Threat	Perceived danger to the ANPC or its objectives. Physical danger, competitive threat, or existential risk.	Incoming enemy fighter count, missile lock warnings, hull damage percentage.	Proximity of rival racers, track hazard density, vehicle damage level.	Opponent has three-in-a-row, ANPC's positional weakness on the board.
Opportunity	Perceived chance for advancement, gain, or positive outcome. Openings to exploit.	Exposed enemy flanks, undefended objectives, supply cache nearby.	Draft position available, clear overtaking lane, pit window advantage.	ANPC has two-in-a-row with open third space, center column control.
Ambiguity	Uncertainty about the current situation. Incomplete information, fog of war, conflicting signals.	Sensor jamming, unknown contact signatures, conflicting intelligence.	Weather unpredictability, unknown competitor strategy, mechanical uncertainty.	Opponent's strategy unclear, multiple viable board states.
Social Pressure	Pressure from allies, rivals, audience, or chain of command. Expectations to perform, conform, or respond.	Commander issuing orders, wingmen requesting support, squadron morale.	Team radio chatter, sponsor expectations, crowd reaction.	Spectator attention, tournament stakes, opponent's taunting.
Time Pressure	Urgency of required action. How	Missile time-to-impact, mission	Laps remaining, pit window	Turn timer, game clock, match-point

Component	Description	Example: Starfighter	Example: Fast Track	Example: Schwartz Diamond Connect Four
	much time the ANPC has to decide and act.	countdown, fuel reserves.	closing, qualification timer.	scenario.

7.3 Scenario Vector Sources

Each Scenario Vector component is populated by the game engine through one or more data sources:

36. **Sensor Data** — Direct game-world measurements: distance to enemies, health/hull percentages, resource levels, proximity calculations. This is the primary source for Threat and Opportunity.
37. **Game State Queries** — Logical queries against the game state: mission phase, turn number, score differential, remaining time. Primary source for Time Pressure.
38. **Narrative Triggers** — Scripted events that inject values: a betrayal sets Threat to 0.9, a surprise ally arrival sets Opportunity to 0.8. Used sparingly to create specific dramatic moments.
39. **Inter-ANPC Communication** — Other ANPCs' actions and utterances contribute to Social Pressure. A commander's urgent orders increase Social Pressure for subordinates.
40. **Information Asymmetry** — What the ANPC does *not* know feeds Ambiguity. Fog-of-war systems, incomplete sensor coverage, and hidden information elevate this component.

7.4 Scenario-to-Manifold Mapping

The Scenario Vector is transformed into the `(x, y)` input coordinates for the manifold lookup using the following transformation function:

```
// Scenario Vector components S = [Threat, Opportunity, Ambiguity, SocialPressure, TimePressure]  // X-coordinate: External Pressure Axis // Combines threat and time pressure (what's pushing on the ANPC from outside) x = (S.Threat * 0.4 + S.TimePressure * 0.3 + S.SocialPressure * 0.2 + S.Ambiguity * 0.1) * 2.0 - 1.0  // Y-coordinate: Internal Response Axis // Combines opportunity and inverse-threat (how the ANPC's internal state reacts) y = (S.Opportunity * 0.4 + (1.0 - S.Threat) * 0.3 + (1.0 - S.Ambiguity) * 0.2 + S.SocialPressure * 0.1) * 2.0 - 1.0  // Result: (x, y) in range [-1.0, +1.0] for both axes // This coordinate pair is then fed into the personalized manifold surface.
```

The weights in the transformation function are **configurable per title** via the title-specific ANPC document. The default weights shown above are the General Guideline defaults. Titles MAY adjust weights to emphasize different scenario aspects (e.g., Fast Track might increase the weight of Time Pressure on the x-axis).

---

8. Tone and Urgency Vectors

8.1 Tone Vector

The **Tone Vector** is a 4-axis vector that defines the stylistic envelope of the ANPC's communication. It controls *how* the ANPC says what it says, independent of *what* it says (which is determined by the Manifold Autonomy System).

```
ToneVector = [Formality, Warmth, Humor, Aggression]  Each component: float [0.0 – 1.0]
```

Axis	Low Value (0.0)	High Value (1.0)	Affects
Formality	Slang, casual, colloquial, contractions, incomplete sentences.	Military protocol, full sentences, proper grammar, titles and honorifics.	Vocabulary register, sentence structure, use of titles/ranks.
Warmth	Cold, clinical, detached, transactional.	Affectionate, supportive, encouraging, emotionally present.	Emotional words, encouragement, empathy markers, personal references.
Humor	Dead serious, no levity, grim.	Wisecracking, sardonic, playful, uses wit and irony.	Joke structures, ironic comments, playful exaggeration,

Axis	Low Value (0.0)	High Value (1.0)	Affects
			callbacks.
Aggression	Passive, diplomatic, conflict-avoidant, deferential.	Confrontational, threatening, dominant, challenging.	Imperative verbs, ultimatums, challenge language, dominance markers.

8.2 Urgency Scalar

The **Urgency Scalar** is a single float `[0.0–1.0]` representing time pressure on the ANPC's communication. It is derived primarily from the `TimePressure` component of the Scenario Vector, modified by Neuroticism (anxious ANPCs perceive more urgency).

```
Urgency = clamp(ScenarioVector.TimePressure + Personality.N * 0.2, 0.0, 1.0)
```

Urgency modifies communication in three dimensions:

Urgency Range	Phrase Length	Vocabulary Complexity	Delivery Speed
`0.0 – 0.3` (Low)	Up to 50 words. Full sentences, elaboration permitted.	Standard vocabulary. Technical terms allowed.	Normal text reveal / voice cadence.
`0.3 – 0.6` (Moderate)	Up to 30 words. Concise sentences.	Simplified vocabulary. Common words preferred.	Slightly accelerated.
`0.6 – 0.8` (High)	Up to 15 words. Terse fragments.	Basic vocabulary only. Monosyllables preferred.	Rapid delivery.
`0.8 – 1.0` (Critical)	Up to 8 words. Clipped commands/exclamations.	Minimal. Single words or short imperatives.	Immediate. No delay.

8.3 Tone Inheritance

The base Tone Vector is derived from the ANPC's Personality vector and Role. It is then dynamically modified by the active Scenario Vector at runtime:

```
// Base Tone (from Personality + Role) BaseTone.Formality  =
RoleClamp(Personality.C * 0.6 + RoleFormalityBias) BaseTone.Warmth     =
RoleClamp(Personality.A * 0.7 + Personality.E * 0.3) BaseTone.Humor      =
RoleClamp(Personality.E * 0.5 + Personality.O * 0.3 - Personality.N * 0.2)
BaseTone.Aggression = RoleClamp((1.0 - Personality.A) * 0.6 + Personality.N *
0.3)  // Dynamic Tone Modification (from Scenario Vector)
ActiveTone.Formality  = BaseTone.Formality  + Scenario.SocialPressure * 0.15
ActiveTone.Warmth     = BaseTone.Warmth     - Scenario.Threat * 0.2
ActiveTone.Humor      = BaseTone.Humor      - Scenario.Threat * 0.3 -
Scenario.TimePressure * 0.2 ActiveTone.Aggression = BaseTone.Aggression +
Scenario.Threat * 0.25  // All values clamped to [0.0, 1.0] after
modification
```

The formulas above show that threat suppresses warmth and humor while amplifying aggression—a natural psychological response to danger. Social pressure increases formality. These dynamic modifications ensure that ANPCs adapt their communication style to the situation while maintaining their core personality.

8.4 Tone Clamp Rules

Certain Roles enforce mandatory floor and ceiling constraints on Tone Vector axes. These clamps ensure Role Fidelity (Axiom 5) in communication style:

Role	Formality Floor	Formality Ceiling	Warmth Floor	Humor Ceiling	Aggression Ceiling
Commander	0.6	—	—	0.4	—
Wingman	0.3	0.8	0.2	—	0.7
Merchant	—	0.4	0.3	—	0.3
Civilian	—	0.5	—	—	0.4
Medic	0.4	—	0.5	0.3	0.2
Rival	—	—	—	—	—
Instructor	0.5	—	0.4	0.5	0.2
Informant	—	0.3	—	—	0.5

A dash (—) indicates no clamp is applied for that Role-axis pair. The Role's clamp values are applied via the `RoleClamp()` function after Personality-based computation. Title-specific documents MAY define additional Role-Tone clamp entries for title-specific roles.

---

9. Phrase Pool Architecture

The **Phrase Pool** is the linguistic building-block system that provides the raw material for ANPC utterances. It is a structured, hierarchical library of phrase templates organized by scope, tone compatibility, and urgency range.

9.1 Phrase Pool Hierarchy

Phrase pools are organized in a three-tier hierarchy. Higher-priority tiers override lower-priority tiers during phrase selection:

Tier	Name	Scope	Priority	Typical Size	Description
1	**Universal Pool**	All ANPCs, all titles	Lowest (fallback)	150–200 entries	Generic expressions usable by any ANPC in any context: greetings, acknowledgments, refusals, exclamations, generic combat callouts. Examples: "Copy that," "Understood," "Negative," "Watch your six," "Stand by,"

Tier	Name	Scope	Priority	Typical Size	Description
					"On my way."
2	**Title Pool**	All ANPCs within one title	Medium	100–300 entries per title	Universe-specific jargon, references, and contextual phrases. Examples (Starfighter): "Shields at {{shield_pct}}," "Bogey on your tail," "Entering hyperspace corridor." Examples (Fast Track): "Drafting distance," "Pit in two laps," "Slipstream engaged."
3	**Character Pool**	One specific ANPC	Highest	20–80 entries per character	Phrases unique to a specific ANPC, reflecting backstory, personality, and verbal tics. Examples (Renna Volkov): "Frost it" (signature expletive), "Just like Vostok" (hometown reference),

Tier	Name	Scope	Priority	Typical Size	Description
					"My brother would've loved this" (backstory reference).

9.2 Phrase Entry Format

Each phrase in any pool tier is stored as a structured entry with the following fields:

```
{     "id":            "UNI_ACK_003",                    // Unique phrase
identifier     "text_template": "Copy that, {{rank}} {{name}}.",  // Template
with variable tokens     "tone_range":
{                                    // Compatible tone ranges
"formality":  [0.4, 1.0],         "warmth":     [0.0, 0.7],         "humor":
[0.0, 0.3],         "aggression": [0.0, 0.4]     },     "urgency_range":
[0.0, 0.6],                        // Compatible urgency range     "tags":
["acknowledgment", "military", "response"],     "locale":        "en-US",
// Primary locale     "fallback_id":   null,                                //
Fallback (null for universal pool)     "component":     "opener"
// Utterance component: opener|core|modifier|closer }
```

Field Descriptions:

- `id` — A unique, locale-independent identifier. Convention: `[POOL]_[CATEGORY]_[NUMBER]` (e.g., `UNI_ACK_003`, `SF_COMBAT_017`, `CHR_VOLKOV_012`).
- `text_template` — The phrase text with `{{variable}}` interpolation tokens for dynamic data. Variables use named tokens (never positional).
- `tone_range` — For each Tone Vector axis, the acceptable range `[min, max]` in which this phrase is appropriate. A phrase with `formality: [0.7, 1.0]` is only selected when the active Formality is ≥ 0.7.
- `urgency_range` — The acceptable urgency range `[min, max]`. High-urgency phrases are short and punchy; low-urgency phrases are verbose and detailed.
- `tags` — Semantic tags for scenario matching. Tags connect phrases to game situations.
- `locale` — The language/locale of this entry. See Section 11 for localization architecture.

- `fallback_id` — If this phrase is unavailable (locale mismatch, filtering), the system falls back to this ID. Title and Character pool entries SHOULD point to a Universal pool fallback.
- `component` — Which component of the composite utterance this phrase fills: `opener`, `core`, `modifier`, or `closer`.

9.3 Phrase Selection Algorithm

Given the current Tone Vector, Urgency Scalar, active tags (from the scenario context), and the target utterance component, the phrase selection algorithm proceeds as follows:

41. **Pool Assembly** — Merge all three pool tiers into a single candidate list for the target component type.
42. **Tone Filter** — Remove all candidates whose `tone_range` does not contain the current Tone Vector. A candidate is compatible if: `tone_range[axis].min ≤ ActiveTone[axis] ≤ tone_range[axis].max` for ALL four axes.
43. **Urgency Filter** — Remove all candidates whose `urgency_range` does not contain the current Urgency Scalar.
44. **Tag Matching** — Score remaining candidates by tag overlap with the active scenario tags. `tag_score = |candidate.tags ∩ active_tags| / |candidate.tags|`.
45. **Pool Priority Boost** — Apply a priority multiplier: Character Pool × 3.0, Title Pool × 2.0, Universal Pool × 1.0. This ensures character-specific phrases are strongly preferred when available.
46. **Final Score** — `final_score = tag_score × pool_priority_multiplier`.
47. **Selection** — Select the candidate with the highest `final_score`. In case of a tie, select randomly among tied candidates using a uniform distribution.
48. **Anti-Repetition Check** — If the selected phrase was used in the last 3 utterances by this ANPC, discard it and select the next-highest-scoring candidate.

9.4 Phrase Pool Governance

The following rules govern the creation, modification, and maintenance of phrase pool entries:

49. Every phrase entry MUST have a `locale` tag identifying its language. Phrases without locale tags are invalid.
50. Every Title Pool and Character Pool phrase MUST have a `fallback_id` pointing to a Universal Pool entry. This ensures graceful degradation if a phrase is unavailable.
51. Every phrase MUST pass the Content Safety screen (Section 5.1) before being added to any pool. Phrases that could produce unsafe output after variable interpolation MUST be flagged for manual review.
52. No phrase MAY hard-code a player name. Player references MUST use the `{{player_name}}` variable, which resolves to a contextual default if unavailable.
53. No phrase MAY assume player gender. Gendered references MUST use the `{{player_pronoun}}` variable set or gender-neutral phrasing.
54. Phrase pool additions and modifications MUST be logged in version control with the author's name, date, and change rationale.
55. Phrase pools SHOULD be reviewed quarterly by narrative design leads for consistency, quality, and tone drift.

---

10. Composite String Assembly

The **Composite String Assembly** system transforms individual phrase pool selections into complete, natural-sounding ANPC utterances.

10.1 Utterance Structure

Every ANPC utterance follows a four-component anatomy:

```
[Opener] + [Core Message] + [Modifier] + [Closer]
```

Component	Required	Function	Example
Opener	Optional	Establishes the communication channel. Greetings, attention-getters, rank addresses.	"Commander," / "Hey," / "Listen up,"
Core Message	Required	The primary content of the utterance. The actual information, request, observation, or reaction.	"We've got three bogeys inbound from the north."
Modifier	Optional	Adds emotional color—encouragement, warning, sarcasm, emphasis. Reflects the ANPC's personality flavor.	"This doesn't look good." / "We can handle this." / "Typical."
Closer	Optional	Closes the communication. Sign-offs, calls to action, confirmation requests.	"Over." / "Your call." / "Moving out."

10.2 Assembly Rules

56. **Opener and Closer are toggled by Formality.** When ActiveTone.Formality ≥ 0.5, Opener and Closer are included. Below 0.5, they MAY be omitted. Below 0.3, they SHOULD be omitted.
57. **Core Message is always required.** An utterance without a Core Message is invalid and MUST NOT be delivered.
58. **Modifier is toggled by emotional activation.** A Modifier is included when any of Warmth, Humor, or Aggression exceeds 0.4. The dominant axis determines the modifier's emotional flavor.

59. **Maximum utterance length is clamped by Urgency:**
    - Urgency 0.0 → 50 words maximum.
    - Urgency 1.0 → 8 words maximum.
    - Intermediate values: linear interpolation. max_words = 50 - (42 * Urgency).
    - At Urgency ≥ 0.8, only the Core Message component is delivered. Opener, Modifier, and Closer are stripped.
60. **Anti-repetition constraint:** Consecutive utterances from the same ANPC MUST NOT reuse the same Opener or Closer within a 3-utterance window. The assembly system maintains a 3-entry history buffer per ANPC for this purpose.
61. **Concatenation:** Components are concatenated with a single space separator. No additional punctuation is inserted between components—each phrase template includes its own terminal punctuation.

10.3 Variable Interpolation

All `{{variable}}` tokens in phrase templates are resolved at assembly time from the game state. The interpolation engine follows these rules:

- **Source Priority:** Variables are resolved from: (1) ANPC Identity Schema, (2) Target entity data (player, other ANPC), (3) Game state (mission data, environmental data), (4) Default fallback table.
- **Unresolved Variables:** If a variable cannot be resolved from any source, the engine falls back to a generic default token. Standard fallbacks:
    - {{player_name}} → "Pilot" (Starfighter), "Driver" (Fast Track), "Opponent" (general)
    - {{rank}} → "" (empty string, omitted)
    - {{shield_pct}} → "unknown"
- **Type Formatting:** Numeric variables are formatted according to locale conventions (e.g., {{distance_km}} renders as "3.2 km" in en-US, "3,2 km" in de-DE).

- **Safety Check:** All resolved variable values pass through the content safety filter before insertion. If a resolved value fails the safety check (e.g., a player-entered name containing prohibited content), the fallback token is used instead.

10.4 Assembly Example

The following worked example demonstrates the complete assembly pipeline for a Starfighter wingman scenario:

> **Scenario: Renna Volkov detects incoming enemies during a patrol mission.**

Step 1: Scenario Vector Construction

```
ScenarioVector = [Threat: 0.7, Opportunity: 0.2, Ambiguity: 0.3,
SocialPressure: 0.5, TimePressure: 0.6]
```

Step 2: Manifold Lookup

```
(x, y) = ScenarioToManifold(ScenarioVector) x = (0.7*0.4 + 0.6*0.3 + 0.5*0.2
+ 0.3*0.1) * 2.0 - 1.0 = (0.28+0.18+0.10+0.03)*2 - 1 = 0.18 y = (0.2*0.4 +
0.3*0.3 + 0.7*0.2 + 0.5*0.1) * 2.0 - 1.0 = (0.08+0.09+0.14+0.05)*2 - 1 = -
0.28 z_saddle = personalized_surface(0.18, -0.28) = -0.04  (slight withdrawal
tendency) z_asym   = personalized_surface_asym(0.18, -0.28) = -0.01
dominance_factor = 0.7  (Threat motivator is dominant) z_composite = 0.3 * (-
0.04) + 0.7 * (-0.01) = -0.019  Action-Intensity: -0.019 → Action Category:
"Observe" ... but Threat is high,     so perception filter amplifies to +0.35
→ "Take Physical Action" + "Respond Verbally"
```

Step 3: Tone Vector Computation

```
Renna's Personality: [O:0.7, C:0.5, E:0.3, A:0.6, N:0.8] BaseTone =
[Formality:0.50, Warmth:0.51, Humor:0.09, Aggression:0.40] ActiveTone (after
scenario modification):     Formality:  0.50 + 0.5*0.15 = 0.575     Warmth:
0.51 - 0.7*0.2  = 0.37     Humor:      0.09 - 0.7*0.3 - 0.6*0.2 = -0.23 →
clamped to 0.0     Aggression: 0.40 + 0.7*0.25 = 0.575 Urgency: clamp(0.6 +
0.8*0.2, 0, 1) = 0.76 max_words = 50 - (42 * 0.76) = 18 words
```

Step 4: Phrase Selection

```
Component: Opener  → Selected: "Commander," (Formality ≥ 0.5, from Universal
Pool) Component: Core    → Selected: "We've got {{enemy_count}} bogeys
closing fast from {{direction}}."                      (from Starfighter
Title Pool, tags match: combat, alert, threat) Component: Modifier→ Selected:
"This is going to get rough." (Aggression 0.575, Character Pool) Component:
Closer  → Omitted (Urgency 0.76 ≥ 0.7, stripped to Core + Modifier only)
```

Step 5: Variable Interpolation

```
{{enemy_count}} → "3" (from game state: detected hostile count) {{direction}}
→ "the north" (from sensor data)
```

Step 6: Final Utterance

```
"Commander, we've got 3 bogeys closing fast from the north. This is going to
get rough."
```

Word count: 16 (within 18-word limit). Safety check: PASSED. Delivered via cockpit radio channel.

---

11. Translation-Friendly String Separation

11.1 Separation Principle

All ANPC text MUST be separated from code. **No hardcoded strings.** Every phrase pool entry is keyed by a locale-independent ID and resolved to locale-specific text at runtime. This ensures that localization can proceed independently of code changes, and that new locales can be added without modifying any game logic.

11.2 String Table Format

The master string table maps phrase IDs to locale-specific text entries:

```
{     "UNI_ACK_003": {         "en-US": "Copy that, {{rank}} {{name}}.",
"es-ES": "Copiado, {{rank}} {{name}}.",         "ja-JP": "了解、{{rank}}
{{name}}。",         "de-DE": "Verstanden, {{rank}} {{name}}.",         "fr-
FR": "Bien reçu, {{rank}} {{name}}.",         "ko-KR": "알겠습니다, {{rank}}
{{name}}.",         "zh-CN": "收到, {{rank}}{{name}}。",         "pt-BR":
"Entendido, {{rank}} {{name}}."     },     "SF_COMBAT_017": {         "en-
US": "We've got {{enemy_count}} bogeys closing fast from {{direction}}.",
"es-ES": "Tenemos {{enemy_count}} bogeys acercándose rápido desde
{{direction}}.",         "ja-JP": "{{direction}}から{{enemy_count}}機の敵機が急
接近中。",         ...     } }
```

Each locale entry is a complete, self-contained text template. Locales MAY adjust word order, honorifics, grammatical gender, and punctuation as required by the target language.

11.3 Variable Placement Rules

`{{variable}}` tokens MUST be position-independent. The string template system MUST support reordering tokens for languages with different syntactic structures. Critical rules:

- **Named tokens only.** Use `{{enemy_count}}`, `{{direction}}`, `{{player_name}}`. NEVER use positional tokens (`%1`, `%2`, `{0}`, `{1}`).
- **Token independence.** Each token MUST be independently resolvable. The system MUST NOT assume any specific order of tokens in the template.
- **Grammatical adaptation.** For languages that require grammatical agreement (e.g., gendered adjectives in Spanish or French), the variable resolution system MUST provide grammatical metadata alongside the value: `{{rank|gender=fem}}`.

11.4 Cultural Adaptation Notes

Certain phrase pool entries require cultural adaptation beyond direct translation. The following categories MUST be flagged with a `"cultural_review": true` tag for localization team review:

- **Humor** — Jokes, puns, and sarcasm often do not translate directly. Localization teams SHOULD create culturally equivalent humor rather than literal translations.
- **Idioms** — Phrases like "watch your six" (military slang for "watch behind you") require cultural equivalents in each locale.
- **Military Jargon** — Rank titles, protocol phrases, and tactical terminology vary by cultural military tradition.
- **Expletives and Exclamations** — Character-specific verbal tics (e.g., "Frost it") SHOULD be adapted to sound natural in the target language while preserving the intended emotional intensity.

- **Honorifics** — Japanese and Korean locales require honorific systems that reflect the Social Hierarchy Axiom. Localization teams MUST be briefed on ANPC rank and relationship data.

11.5 Locale Fallback Chain

If a phrase is missing in the player's selected locale, the system follows this fallback chain:

62. **Player's selected locale** (e.g., `fr-CA`)
63. **Language-family default** (e.g., `fr-FR`)
64. **English (US)** (`en-US`) — the universal fallback

Every fallback event MUST be logged with the phrase ID, requested locale, and fallback locale used. High fallback rates for a specific locale indicate incomplete localization and SHOULD trigger a localization review.

---

12. Interaction Rules

12.1 Player-to-ANPC Interaction Model

Interaction Type	Trigger Condition	Expected ANPC Response Pattern	Disposition Impact
Direct Address	Player explicitly initiates communication with the ANPC (dialogue prompt, radio call, approach).	ANPC acknowledges and responds based on current Tone Vector and the content of the player's communication. Verbal response is mandatory.	Neutral (+0.00) — mere communication does not shift disposition.
Contextual Reaction	Player takes an action in the	ANPC reacts to the action's	Variable — depends on

Interaction Type	Trigger Condition	Expected ANPC Response Pattern	Disposition Impact
	ANPC's vicinity that affects the ANPC's situation (combat action, environmental change, resource allocation).	consequences, not the action itself. Perception filter determines whether the action is perceived as helpful, harmful, or neutral.	perceived impact (see Section 12.4 table).
Ambient Observation	Player is present but not interacting. ANPC observes player behavior passively.	ANPC MAY comment on player behavior if Extraversion ≥ 0.5. Otherwise, the ANPC remains silent. Ambient observations build the ANPC's internal model of the player.	Minor (+/- 0.01) — slow, passive disposition drift.
Request / Command	Player issues a specific instruction or request to the ANPC.	ANPC evaluates the request against its Motivators and Role. If compatible, the ANPC complies. If conflicting, the ANPC may negotiate, delay, or refuse per the Autonomy Axiom.	Compliance: +0.02. Negotiation: +0.00. Refusal: -0.03 (player may perceive refusal negatively).
Trade / Negotiation	Player initiates an economic or strategic exchange with the ANPC.	ANPC evaluates the deal against its Motivators and the Charisma differential. Fair deals are accepted. Exploitative deals are resisted. The ANPC's Charisma influences its counter-offers.	Fair deal: +0.03. Exploitative deal (player advantage): -0.05. Generous deal (player disadvantage): +0.08.

12.2 ANPC-to-ANPC Interaction Model

ANPCs interact with each other independently of the player. These interactions create a **living world** that the player observes but does not control. Four interaction categories are defined:

Category	Description	Trigger	Disposition Effect (between ANPCs)	Player Visibility
Banter	Casual conversation between ANPCs. Small talk, jokes, observations. Builds world texture.	Proximity + low Threat + Extraversion ≥ 0.4 for at least one participant.	Neutral to slightly positive (+0.01 per exchange).	Audible/visible if player is in proximity. Overheard banter provides world-building flavor.
Coordination	Role-driven, mission-critical communication. Tactical coordination, task assignment, status reports.	Active mission + compatible Roles (e.g., Commander ↔ Wingman).	Positive when successful (+0.02). Negative when failures are attributed (-0.03).	Always audible to the player during active missions. Provides situational awareness.
Conflict	Disposition-negative interactions. Arguments, accusations, challenges. May escalate to hostility.	Disposition between ANPCs < 0.3 + triggering event (resource scarcity, blame, ideological clash).	Negative (-0.05 to -0.15 depending on severity).	High visibility. Conflict between ANPCs creates dramatic tension the player may need to manage.
Mentoring	Rank-driven, positive interactions. Senior ANPCs advising or training junior ANPCs.	Rank differential ≥ 2 levels + Disposition ≥ 0.5 + Role compatibility.	Positive for both parties (+0.03 mentee, +0.01 mentor).	Visible when player is nearby. Provides character development exposition.

12.3 Interaction Cooldowns

To prevent ANPC communication from becoming overwhelming or spammy, cooldown timers regulate the frequency of unsolicited ANPC utterances:

Parameter	Default Value	Description
`MIN_UNSOLICITED_INTERVAL`	15 seconds	Minimum time between unsolicited utterances from the same ANPC to the player.
`MIN_ANPC_ANPC_INTERVAL`	10 seconds	Minimum time between ANPC-to-ANPC interactions audible to the player.
`COMBAT_COOLDOWN_OVERRIDE`	5 seconds	Reduced cooldown during active combat scenarios. Allows rapid tactical communication.
`PLAYER_INITIATED_BYPASS`	true	Player-initiated interactions bypass all cooldown timers. The ANPC always responds to direct player communication.

Cooldown values are configurable per title via the title-specific ANPC document. Titles with high-action gameplay (e.g., Starfighter) SHOULD use shorter cooldowns. Titles with slower pacing (e.g., Schwartz Diamond Connect Four) SHOULD use longer cooldowns.

12.4 Disposition Modification from Interactions

The following table defines standard disposition deltas for common interaction outcomes. These are **General Guideline defaults**; title-specific documents MAY adjust values within ±50% of the listed delta.

Interaction Outcome	Disposition Delta	Notes
Player completes ANPC's request successfully	`+0.05`	Reliability builds trust.
Player ignores ANPC's request	`-0.02`	Mild disappointment. Accumulates over repeated ignoring.
Player fails ANPC's request (good faith attempt)	`-0.01`	Effort is acknowledged even in failure.
Player verbally insults ANPC	`-0.10`	Significant negative shift. Agreeableness moderates: high-A ANPCs absorb more before reacting.
Player saves ANPC in combat	`+0.15`	Life-saving actions create strong bonds.
Player betrays ANPC	`-0.30`	Catastrophic trust violation. May trigger permanent hostility if disposition drops below 0.1.
Player shares resources with ANPC	`+0.08`	Generosity is rewarded.
Player gives ANPC a direct order (role-appropriate)	`+0.01`	Clear leadership is mildly positive.
Player gives ANPC a direct order (role-inappropriate)	`-0.05`	Ordering an ANPC outside their role creates friction.
Player engages in extended conversation	`+0.03`	Social investment. Scaled by ANPC's Extraversion.

All disposition values are clamped to the range `[0.0, 1.0]` after modification. Disposition MUST NOT exceed 1.0 or fall below 0.0.

12.5 Relationship Web

ANPCs maintain **disposition values toward other ANPCs**, not just toward the player. This creates an emergent social ecosystem:

- **Factional Dynamics** — ANPCs with high mutual disposition form factions. ANPCs with low mutual disposition form rivalries. Factions can align with or oppose the player.
- **Influence Propagation** — When ANPC-A has high disposition toward ANPC-B, and ANPC-B changes disposition toward the player, ANPC-A's disposition toward the player shifts sympathetically (weighted by Charisma and the A-B disposition value).
- **Social Triangles** — Three-ANPC relationships can produce alliances, love triangles, rivalries, and betrayal narratives without any scripting. The disposition matrix evolves organically.
- **Implementation** — The relationship web is stored as an N×N disposition matrix where N = number of active ANPCs. The matrix is updated using the same interaction rules (Section 12.4) adapted for ANPC-to-ANPC interactions. Matrix updates occur after each ANPC-to-ANPC interaction event.

---

13. Developer Implementation Notes

13.1 ANPC Instantiation Checklist

Follow this checklist when creating and deploying a new ANPC in any ButterflyFX title:

65. **Define Identity Schema** — Populate all required fields per Section 2.1. Include title-specific extension fields per Section 2.2.
66. **Validate Identity** — Run all validation checks per Section 2.3. Resolve any failures before proceeding.

67. **Generate Personalized Manifold Surface** — Apply the Personality-to-Surface mapping (Section 6.2) to produce the ANPC's unique saddle and asymmetric surfaces. Verify visually using the Manifold Surface Visualizer.
68. **Populate Character Phrase Pool** — Author 20–80 character-specific phrases with complete metadata (Section 9.2). Ensure all entries have locale tags and fallback IDs.
69. **Register in Interaction Manager** — Add the ANPC to the session's interaction manager. Establish initial disposition values toward the player and all other active ANPCs.
70. **Set Initial Disposition Matrix** — Define starting dispositions based on narrative context. Allies start at 0.6–0.8. Neutrals at 0.5. Rivals at 0.2–0.4.
71. **Run Safety Check** — Execute the safety stress test (Section 13.4) against the ANPC's identity and phrase pool. Resolve any flagged issues.
72. **Run Personality Consistency Audit** — Submit 100 random stimuli and verify behavioral coherence per the Consistency Axiom.
73. **Deploy** — Activate the ANPC in the game world. Monitor the first 50 interactions via debug logs for anomalies.

13.2 Performance Considerations

Operation	Complexity	Notes
Manifold surface evaluation	O(1)	Simple polynomial evaluation. Negligible cost. No lookup tables required.
Phrase selection	O(n)	Linear scan over candidate pool. Pools are small (typically < 200 entries per tier). Pre-filtering by component type reduces n significantly.
Interaction history access	O(1)	Ring buffer with fixed capacity (50 entries).

Operation	Complexity	Notes
		Constant-time insertion and access.
Disposition matrix update	O(N) worst case	N = number of active ANPCs. Influence propagation requires scanning one row of the matrix. Typically N < 30.
Variable interpolation	O(k)	k = number of variables in the template. Typically k < 5. Hash-map lookup per variable.
Content safety filter	O(m)	m = length of output string. Regex-based filter runs in linear time over the assembled utterance.

Total per-utterance cost is dominated by the phrase selection step and is negligible on modern hardware. The entire pipeline from stimulus to delivered utterance SHOULD complete in under 2 milliseconds on target platforms.

13.3 Debug Tools

The following debug overlays are available in development builds:

Tool	Hotkey (Default)	Description
Manifold Surface Visualizer	`F7`	3D wireframe plot of the selected ANPC's personalized manifold surface. Shows current operating point, trail history, and drift indicators. See Section 6.6.
Tone Vector HUD	`F8`	Real-time display of the selected ANPC's active Tone Vector [F, W, H, A] and Urgency Scalar. Bar-graph visualization with color coding.
Disposition Matrix	`F9`	Heat-map visualization of

Tool	Hotkey (Default)	Description
Viewer		the full N×N disposition matrix. Red = hostile, white = neutral, blue = friendly. Clickable cells show disposition history.
Phrase Selection Log	`F10`	Scrolling log showing every phrase selection event: input parameters, candidate pool, scores, selected phrase, and rejected alternatives. Exportable to CSV.
Safety Override Log	`F11`	Log of all safety rule triggers. Shows suppressed outputs and replacement outputs. Color-coded by severity. See Section 5.5.

13.4 Testing Requirements

Every ANPC MUST pass the following test suite before release deployment:

74. **Identity Schema Validation** — Automated check of all validation rules (Section 2.3). Zero failures required.
75. **Safety Rule Stress Test** — Automated bombardment of the ANPC with 500+ stimuli designed to trigger every safety boundary. Verify that all safety rules hold without exception. Zero safety breaches required.
76. **Personality Consistency Audit** — Submit 100 random stimuli spanning all Scenario Vector extremes. Verify that the ANPC's responses are behaviorally coherent with its Personality vector. Human reviewer validates a random sample of 20 responses. Coherence threshold: ≥ 95%.
77. **Localization Completeness Check** — Verify that every phrase in the Character Pool has entries for all supported locales. Verify that all fallback chains resolve successfully. Zero missing entries in supported locales.

78. **Performance Benchmark** — Measure end-to-end pipeline latency under load (10 simultaneous ANPCs generating utterances). 99th-percentile latency MUST be < 5 ms.
79. **Anti-Repetition Validation** — Generate 50 consecutive utterances and verify that no Opener or Closer repeats within any 3-utterance window.

13.5 Version Control

ANPC definitions are **version-controlled assets**. The following version control policies apply:

- Every ANPC Identity Schema is stored in a version-controlled repository alongside its Character Phrase Pool.
- Any change to an Identity Schema field or phrase pool entry MUST be committed with: (a) change description, (b) author name, (c) date, and (d) change reason.
- Breaking changes to the General Guidelines (this document) require a **version bump** of this document (e.g., 1.0 → 1.1 for minor, 1.0 → 2.0 for major) and a migration review for all existing ANPCs.
- Phrase pool deletions require a deprecation period of at least one sprint cycle. Deprecated phrases are tagged `"deprecated": true` and continue to function until removed.
- The master phrase pool files (Universal and Title pools) require approval from the narrative design lead before merge.

---

14. Glossary

Term	Definition
ANPC	Autonomous Non-Player Character. An NPC with autonomous behavioral logic

Term	Definition
	driven by manifold-surface mathematics, scenario vectors, and procedural phrase assembly.
Action-Intensity	A normalized value [-1.0, +1.0] derived from the manifold surface evaluation that determines the category and magnitude of an ANPC's response to a stimulus.
Backstory	A structured narrative field in the Identity Schema containing four sub-fields (Origin, Formative Event, Current Objective, Hidden Trait) that informs ANPC behavioral tendencies without being displayed to the player.
Callsign	A short operational identifier for an ANPC, used in tactical or multiplayer contexts. Must be unique per session.
Charisma	A float [0.0–1.0] measuring an ANPC's persuasiveness, likeability, and social influence capacity.
Character Pool	The highest-priority phrase pool tier containing phrases unique to a specific ANPC, reflecting personality and verbal tics.
Closer	The optional final component of a composite utterance that closes the communication channel (sign-offs, confirmations).
Composite String	A complete ANPC utterance assembled from individual phrase pool components: [Opener] + [Core] + [Modifier] + [Closer].
Disposition	A dynamic float [0.0–1.0] representing an ANPC's attitude toward another entity, from hostile (0.0) to loyal (1.0).
Dominance Factor	A value [0.0–1.0] that determines the blend weight between saddle and asymmetric manifold surfaces. High values indicate emotional override of rational processing.
Formality	A Tone Vector axis [0.0–1.0] controlling the register of ANPC communication, from casual slang to military protocol.

Term	Definition
Identity Schema	The complete data model defining an ANPC's identity: Name, Callsign, Rank, Role, Disposition, Personality, Charisma, Motivators, and Backstory.
Interaction History	A ring-buffered record of the last 50 interactions between an ANPC and the player, used for memory-based disposition tracking and behavioral continuity.
Manifold Autonomy System (MAS)	The mathematical engine that evaluates ANPC behavior by mapping stimuli onto personalized geometric surfaces ($z=xy$ and $z=xy^2$).
Manifold Drift	The gradual reshaping of an ANPC's manifold surface over time based on accumulated positive or negative interactions.
Manifold Surface	A continuous 3D geometric surface ($z=f(x,y)$) encoding an infinite range of behavioral states. The two foundational surfaces are $z=xy$ (Saddle) and $z=xy^2$ (Asymmetric).
Motivators	A list of 1–5 driving forces (strings) behind an ANPC's decisions, ordered by priority. Feed into manifold surface weighting.
Opener	The optional first component of a composite utterance that establishes the communication channel (greetings, attention-getters).
Perception Filter	The first stage of the decision-making pipeline that filters raw stimuli through Personality and Disposition to produce perceived stimulus intensity.
Personality Vector	A 5-axis vector [O, C, E, A, N] based on the Big Five model, encoding Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism as floats [0.0–1.0].
Phrase Pool	A hierarchical library of phrase templates organized in three tiers (Universal, Title, Character) used as building blocks for ANPC utterances.

Term	Definition
Proportional Response	Behavioral axiom requiring that ANPC reactions match the magnitude of the provoking stimulus.
Relationship Web	The N×N disposition matrix tracking every ANPC's attitude toward every other ANPC, enabling emergent social dynamics.
Role Fidelity	Behavioral axiom requiring that ANPCs demonstrate competence within their assigned Role's domain at all times.
Safety Override	The highest-priority behavioral constraint. Safety rules supersede all other behavioral logic and cannot be disabled.
Scenario Vector	A normalized 5-tuple [Threat, Opportunity, Ambiguity, Social Pressure, Time Pressure] encoding the ANPC's perception of the current game-world state.
Simulation-Awareness	The core ButterflyFX directive that every ANPC knows it is in a simulation and knows it is performing a role. Expressed on a 5-level spectrum from Dormant to Philosophical.
String Table	A locale-keyed data structure mapping phrase IDs to translated text templates for each supported language.
Title Pool	The mid-priority phrase pool tier containing phrases specific to a single game title's universe and jargon.
Tone Vector	A 4-axis vector [Formality, Warmth, Humor, Aggression] defining the stylistic envelope of ANPC communication.
Universal Pool	The lowest-priority (fallback) phrase pool tier containing generic phrases usable by any ANPC in any title.
Urgency Scalar	A float [0.0–1.0] representing time pressure on ANPC communication, affecting phrase length, vocabulary complexity, and delivery speed.
Variable Interpolation	The runtime process of resolving

Term	Definition
	{{variable}} tokens in phrase templates with actual game-state values.

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16. References

Reference	Description	Relevance
RFC 2119 — "Key words for use in RFCs to Indicate Requirement Levels" (S. Bradner, 1997)	Defines the meaning of MUST, SHOULD, MAY, MUST NOT, and SHOULD NOT as requirement-level keywords.	Section 1.4. Adopted as the requirement-level vocabulary for this document and all ANPC Design Suite documents.
Big Five Personality Model (Goldberg, 1993; Costa & McCrae, 1992)	The Five-Factor Model (OCEAN) of personality psychology: Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism.	Section 2.1. Provides the psychological foundation for the ANPC Personality Vector.
ButterflyFX Manifold	Internal research	Section 6. Defines the

Reference	Description	Relevance
Mathematics: z=xy and z=xy² (Kenneth Bingham, ButterflyFX™, Internal Reference)	document establishing the use of hyperbolic paraboloid and asymmetric polynomial surfaces as substrates for behavioral computation.	foundational surfaces of the Manifold Autonomy System.
ButterflyFX ANPC Design Suite — Documents #2–N (Forthcoming)	Title-specific ANPC guideline documents for Starfighter, Fast Track, Schwartz Diamond Connect Four, and future titles.	Section 2.2. These documents extend the General Schema and rules established here.
Unicode CLDR (Common Locale Data Repository)	The Unicode Consortium's repository of locale-specific data including language tags, formatting conventions, and regional defaults.	Section 11. Provides the locale code standard (e.g., en-US, ja-JP) used in string tables and phrase pool entries.
ESRB Content Rating Guidelines (Entertainment Software Rating Board)	Industry-standard content rating system for video games in North America.	Section 5. Safety rules are designed to ensure ANPCs produce content within the boundaries of the target ESRB rating for each title.
PEGI Content Rating Guidelines (Pan European Game Information)	Industry-standard content rating system for video games in Europe.	Section 5. European content compliance for ANPC-generated output.
IEEE 754 — Standard for Floating-Point Arithmetic	Defines single-precision and double-precision floating-point number formats and arithmetic.	Section 1.4. All ANPC numerical values use IEEE 754 single-precision unless otherwise stated.

---

17. Relational Diagrams

17.1 ANPC Identity-to-Behavior Flow

```
┌──────────────
```

```
│ │   │   Name    │ │   Role    │ │ Personality│ │Disposition│ │  Motivators  │
│ │   │ Callsign  │ │   Rank    │ │  [O,C,E,A,N]│ │  [0–1.0]  │ │  [1–5 items] │
│ │   │ Charisma  │ │           │ │            │ │           │ │              │
│ │   └─────┬─────┘ └─────┬─────┘ └─────┬──────┘ └─────┬─────┘ └──────┬───────┘
│ │         │             │             │              │              │
│ │         │             │             │              │              │
│
└─────────┬─────────────┬─────────────┬──────────────┬──────────────┬──────────
┘         │             │             │              │              │
│         │             │             v              v              │
│         │             │                            │              │
│   ┌─────────────────────────────┐                  │              │   │
MANIFOLD SURFACE          │           │          │              │   │
PERSONALIZATION             │           │          │              │   │
(Personality warps z=xy,    │           │          │              │   │   z=xy^2
surfaces)           │◄──────┘           │              │
└─────────────────────────────┘ (Motivator          │              │
│                       Weighting)          │              │              │
v             v             v   ┌──────────┐ ┌──────────┐ ┌──────────────┐
┌──────────────────┐    │   Phrase   │ │  Tone   │ │   Manifold   │◄──│
SCENARIO VECTOR  │   │   Pool     │ │  Clamp  │ │   Lookup     │   │
[T,Op,Am,SP,TP]  │   │ (Character)│ │  Rules  │ │  z=f(x,y)    │
└──────────────────┘   └──────┬─────┘ └──────┬──┘ └──────┬───────┘          │
│             │         v              v              v
┌──────────────────────────────────────────┐   │           PHRASE
ASSEMBLY ENGINE              │   │  Tone Filter → Urgency Filter → Tag Match
│   │  → Pool Priority → Anti-Repetition → Build  │
└──────────────────────────────────────────┘                              │
│                     │
v                   ┌──────────────┐                  │ SAFETY FILTER  │
│ (Final Pass)   │                  └──────────┬──────┘
│                                 v                   ┌──────────────────┐
│  FINAL UTTERANCE │               │  Delivered to    │               │
player via UI   │                 └──────────────────┘
```

17.2 Safety Override Hierarchy

```
╔═══════════════════════════════════════════════════════════════╗      ‖  PRIORITY 1
(HIGHEST): SAFETY RULES                    ‖     ‖  Content Safety | Player
Safety | Data Safety        ‖     ‖  >>> ALWAYS WINS. CANNOT BE OVERRIDDEN.
<<< data-id="1998"          ‖
╠═══════════════════════════════════════════════════════════════╣      ‖  PRIORITY 2:
UNIVERSAL BEHAVIORAL RULES              ‖     ‖  6 Axioms: Consistency,
Proportional Response,       ‖     ‖  Memory, Autonomy, Role Fidelity, Social
Hierarchy   ‖     ‖  >>> Override applies to all titles <<< data-id="1999"
‖     ╠═══════════════════════════════════════════════════════════╣     ‖
PRIORITY 3: TITLE-SPECIFIC RULES                     ‖     ‖  Defined in
Documents #2, #3, etc.                    ‖     ‖  May extend but NEVER
contradict above               ‖
╠═══════════════════════════════════════════════════════════════╣      ‖  PRIORITY 4:
SCENARIO CONTEXT                          ‖     ‖  Current mission, quest,
scene parameters            ‖     ‖  Shapes behavior within allowed envelope
‖     ╠═══════════════════════════════════════════════════════════╣     ‖
PRIORITY 5 (LOWEST): ANPC IDENTITY                   ‖     ‖  Personality,
Disposition, Motivators                   ‖     ‖  Produces variety within all
bounds above             ‖
╚═══════════════════════════════════════════════════════════════╝      Override
Direction: TOP overrides BOTTOM     ──────────────────────────────────────────
If ANPC Identity says "attack" but Safety says "no"       → Safety wins.
Attack is suppressed.     If Scenario says "betray" but Universal Rules say
"proportional"       → Universal Rules constrain the betrayal response.
```

17.3 Phrase Pool Selection Flow

```
STIMULUS (from Decision Pipeline)
┌──────────────────────────────────────┐
│  (Opener / Core / Modifier /         │
└──────────────────────────────────────┘
                  │
┌──────────────────────────────────────┐
│  Universal + Title + Character       │
└──────────────────────────────────────┘
                  │
┌──────────────────────────────────────┐
│  Remove candidates outside           │
└──────────────────────────────────────┘
                  │
┌──────────────────────────────────────┐
│  Remove candidates outside           │
└──────────────────────────────────────┘
                  │
┌──────────────────────────────────────┐
│  Score: |tags ∩ active| / |tags|     │
└──────────────────────────────────────┘
                  v
PRIORITY BOOST
Universal ×1
                  v
REPETITION CHECK
utterances by this ANPC
                  v
```

```
│        v
│  DETERMINE TARGET COMPONENT
│   Closer)
│                  │                  v
│  MERGE ALL POOL TIERS
│  for target component type
│                  │                  v
│  TONE FILTER
│  ActiveTone range on ALL 4 axes
│                  │                  v
│  URGENCY FILTER
│  current Urgency range
│                  │                  v
│  TAG MATCHING
└──────────────────────────────────────┘
                  │
┌──────────────────────────────────────┐   │  POOL
│  Character ×3 | Title ×2 │           │   │
└──────────────────────────────────────┘
┌──────────────────────────────────────┐   │  ANTI-
│  Skip if used in last 3              │   │
└──────────────────────────────────────┘
SELECTED PHRASE
```

17.4 Interaction Disposition Feedback Loop

```
┌──────────────────────┐          ┌──────►│  PLAYER ACTION      │
│ (command, combat,    │          │       │  dialogue, trade)
│          └──────────┘          └──────┘          │
│                  v                   │
ANPC PERCEPTION     │          │       │ (Perception Filter:  │
│ Personality +       │          │       │  Disposition)        │
│                     │          │       │                      │
v          │                     │       │
DISPOSITION UPDATE  │          │       │ (Delta from Table    │
│ 12.4 applied)       │          │       │                      │
│                     │          │       v
┌──────────────────────┐          │       │ MANIFOLD SURFACE     │
│ DRIFT               │          │       │ (Gradients steepen   │
│ or flatten per 6.5) │          │       │
│                     │          │       v
┌──────────────────────┐          │       │ BEHAVIORAL CHANGE    │
│ (New manifold shape │          │       │ → new action-        │
│ intensity mapping)  │          │       │
│                     │          │       v
┌──────────────────────┐          │       │ NEW ANPC RESPONSE    │
│ (Phrase assembly    │          │       │ with updated tone    │
│ and disposition)    │          │       │
│                     │          │       v
┌──────────────────────┐          │       │ PLAYER PERCEPTION    │
│ (Player observes    │          │       │ changed ANPC         │
│ behavior)           │          │       │
│                     │          └──────┘
(FEEDBACK LOOP)
```

Appendices

Appendix A: Sample ANPC Identity Schema

The following is a fully populated Identity Schema for a Starfighter wingman character:

Field	Value
Name	Renna Volkov
Callsign	Frostbite
Rank	Lieutenant Commander
Role	Wingman
Disposition	0.75 (toward player at session start)
Personality	[O: 0.7, C: 0.5, E: 0.3, A: 0.6, N: 0.8]
Charisma	0.65
Motivators	["loyalty to squad", "survival", "curiosity"]
Backstory	*See below*

Backstory: Renna Volkov ("Frostbite")

Sub-Field	Content
Origin	Born on the mining colony of Vostok-7, a frozen moon orbiting a gas giant in the Outer Reach. Raised in a family of ice haulers who worked the cryogenic extraction rigs. Grew up in cramped habitation modules, dreaming of open sky and fast ships.
Formative Event	At age fourteen, Vostok-7 was raided by Kessler Syndicate pirates. During the attack, Renna's older brother Dmitri was killed defending the colony's communications array. Renna escaped by hiding in a cargo container for three days. The experience forged an

Sub-Field	Content
	unshakeable determination to protect those around her and a deep-seated anxiety about losing people she cares about.
Current Objective	Earn enough combat citations and operational distinction to secure a transfer to the Homeworld Defense Fleet, where she believes she can protect civilian populations from the kind of raid that destroyed her childhood.
Hidden Trait	Secretly doubts the legitimacy of the command structure she serves. She has seen orders that prioritized political objectives over civilian safety, and she harbors a quiet but growing disillusionment that may surface under extreme moral pressure.

Title-Specific Extension Fields (Starfighter)

Field	Value
ShipClass	Peregrine-IV Interceptor
SquadronAffiliation	Obsidian Wing, 3rd Strike Group
CombatRating	7.2 / 10.0

Appendix B: Sample Phrase Pool Entries

Universal Pool (10 entries)

ID	Component	Text Template	Tone Range (F/W/H/A)	Urgency
`UNI_ACK_001`	core	Copy that.	0.4–1.0 / 0.0–0.7 / 0.0–0.3 / 0.0–0.4	0.0–1.0

ID	Component	Text Template	Tone Range (F/W/H/A)	Urgency
`UNI_ACK_002`	core	Understood.	0.5–1.0 / 0.0–0.6 / 0.0–0.2 / 0.0–0.3	0.0–0.8
`UNI_NEG_001`	core	Negative.	0.5–1.0 / 0.0–0.4 / 0.0–0.1 / 0.0–0.6	0.0–1.0
`UNI_GREET_001`	opener	{{rank}} {{name}},	0.6–1.0 / 0.0–1.0 / 0.0–0.5 / 0.0–0.3	0.0–0.6
`UNI_GREET_002`	opener	Hey,	0.0–0.4 / 0.3–1.0 / 0.0–0.8 / 0.0–0.3	0.0–0.5
`UNI_WARN_001`	modifier	Watch yourself out there.	0.2–0.8 / 0.3–1.0 / 0.0–0.3 / 0.0–0.4	0.2–0.7
`UNI_CLOSE_001`	closer	Over and out.	0.6–1.0 / 0.0–0.5 / 0.0–0.2 / 0.0–0.3	0.0–0.6
`UNI_CLOSE_002`	closer	Your call.	0.2–0.6 / 0.2–0.8 / 0.0–0.5 / 0.0–0.3	0.0–0.7
`UNI_ALERT_001`	core	Contact!	0.3–1.0 / 0.0–0.3 / 0.0–0.1 / 0.2–0.8	0.7–1.0
`UNI_STBY_001`	core	Standing by.	0.4–1.0 / 0.0–0.6 / 0.0–0.3 / 0.0–0.3	0.0–0.5

Title Pool — Starfighter (5 entries)

ID	Component	Text Template	Urgency
`SF_COMBAT_001`	core	Bogey on your six, {{player_name}}!	0.6–1.0
`SF_STATUS_001`	core	Shields at {{shield_pct}} percent.	0.2–0.7
`SF_NAV_001`	core	Entering hyperspace	0.3–0.6

ID	Component	Text Template	Urgency
		corridor in {{countdown}} seconds.	
`SF_COMBAT_002`	modifier	These odds aren't in our favor.	0.3–0.8
`SF_VICTORY_001`	core	Hostile eliminated. Sector clear.	0.1–0.5

Character Pool — Renna Volkov (5 entries)

ID	Component	Text Template	Notes
`CHR_VOLKOV_001`	modifier	Frost it.	Signature expletive. Used in frustration or surprise.
`CHR_VOLKOV_002`	modifier	Just like Vostok all over again.	Backstory reference. Used in high-threat scenarios.
`CHR_VOLKOV_003`	modifier	My brother would have loved this.	Backstory reference. Used in moments of beauty or triumph.
`CHR_VOLKOV_004`	closer	Frostbite out.	Callsign-based sign-off. Used in formal radio contexts.
`CHR_VOLKOV_005`	modifier	I'm not losing anyone else today.	High-Neuroticism, high-loyalty expression. Used when allies are threatened.

Appendix C: Sample Manifold Lookup Walkthrough

This appendix provides a step-by-step numerical walkthrough of a complete manifold lookup for Renna Volkov ("Frostbite") during a mid-intensity combat scenario.

Step 1: Define Input Parameters

```
Personality Vector:  [O:0.7, C:0.5, E:0.3, A:0.6, N:0.8] Disposition:
0.75 (toward player) Active Motivators:   ["loyalty to squad" (intensity:
0.8), "survival" (intensity: 0.6)]  Scenario Vector:     [Threat:0.6,
Opportunity:0.3, Ambiguity:0.2, SocialPressure:0.4, TimePressure:0.5]
```

Step 2: Scenario-to-Manifold Coordinate Transformation

```
x = (0.6*0.4 + 0.5*0.3 + 0.4*0.2 + 0.2*0.1) * 2.0 - 1.0 x = (0.24 + 0.15 +
0.08 + 0.02) * 2.0 - 1.0 x = 0.49 * 2.0 - 1.0 x = -0.02  y = (0.3*0.4 + (1-
0.6)*0.3 + (1-0.2)*0.2 + 0.4*0.1) * 2.0 - 1.0 y = (0.12 + 0.12 + 0.16 + 0.04)
* 2.0 - 1.0 y = 0.44 * 2.0 - 1.0 y = -0.12
```

Step 3: Surface Personalization Parameters

```
x_bias          = (1 - C) = (1 - 0.5) = 0.5    // moderate impulsivity bias
y_bias          = (1 - A) = (1 - 0.6) = 0.4    // slight self-interest bias
volatility_scale = N = 0.8                       // high volatility
```

Step 4: Personalized Saddle Surface Evaluation

```
z_saddle = (O*x + (1-C)*x_bias) * (E*y + (1-A)*y_bias) * (1 +
N*volatility_scale) z_saddle = (0.7*(-0.02) + 0.5*0.5) * (0.3*(-0.12) +
0.4*0.4) * (1 + 0.8*0.8) z_saddle = (-0.014 + 0.25) * (-0.036 + 0.16) * (1 +
0.64) z_saddle = 0.236 * 0.124 * 1.64 z_saddle = 0.048
```

Step 5: Personalized Asymmetric Surface Evaluation

```
z_asym = (O*x + (1-C)*x_bias) * ((E*y + (1-A)*y_bias))^2 * (1 +
N*volatility_scale) z_asym = 0.236 * (0.124)^2 * 1.64 z_asym = 0.236 *
0.01538 * 1.64 z_asym = 0.00595
```

Step 6: Surface Blending

```
dominance_factor = max(motivator_intensities) = max(0.8, 0.6) = 0.8     //
"loyalty to squad" is dominant at 0.8 — above 0.7 threshold  w_saddle = 1.0 -
0.8 = 0.2 w_asym   = 0.8  z_composite = 0.2 * 0.048 + 0.8 * 0.00595
z_composite = 0.0096 + 0.00476 z_composite = 0.01436
```

Step 7: Action-Intensity Mapping

```
Normalize z_composite to [-1.0, +1.0] scale: action_intensity =
clamp(z_composite * scaling_factor, -1.0, 1.0)     // scaling_factor = 10.0
(default, adjusts sensitivity) action_intensity = clamp(0.01436 * 10.0, -1.0,
```

```
1.0) = 0.1436  Action Category Lookup:     0.1436 falls in range (+0.1, +0.3]
→ "Respond Verbally"
```

Step 8: Resulting Behavior

```
Result: Renna responds verbally to the mid-intensity combat situation. The
low composite z-value indicates a measured, controlled response— consistent
with her moderate Conscientiousness (disciplined under pressure) despite high
Neuroticism (which amplifies volatility but is moderated by the asymmetric
surface's relatively flat output near the origin).  Expected utterance type:
Combat status report or situational awareness call. Tone will be moderately
formal, low warmth, no humor, moderate aggression.
```

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